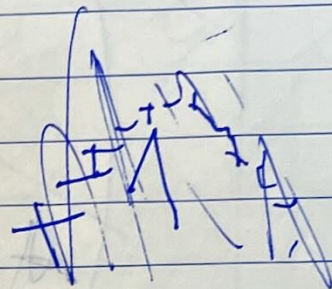


~~SV~~

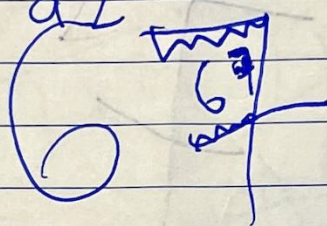
$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = x$$

$$\cos x < \frac{\sin x}{x} < \frac{1}{\cos x}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$



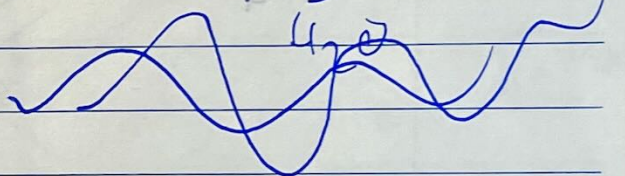
$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



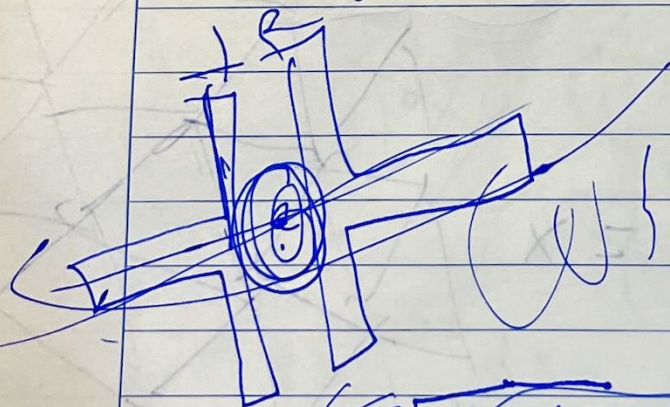
H₂O

H₂O

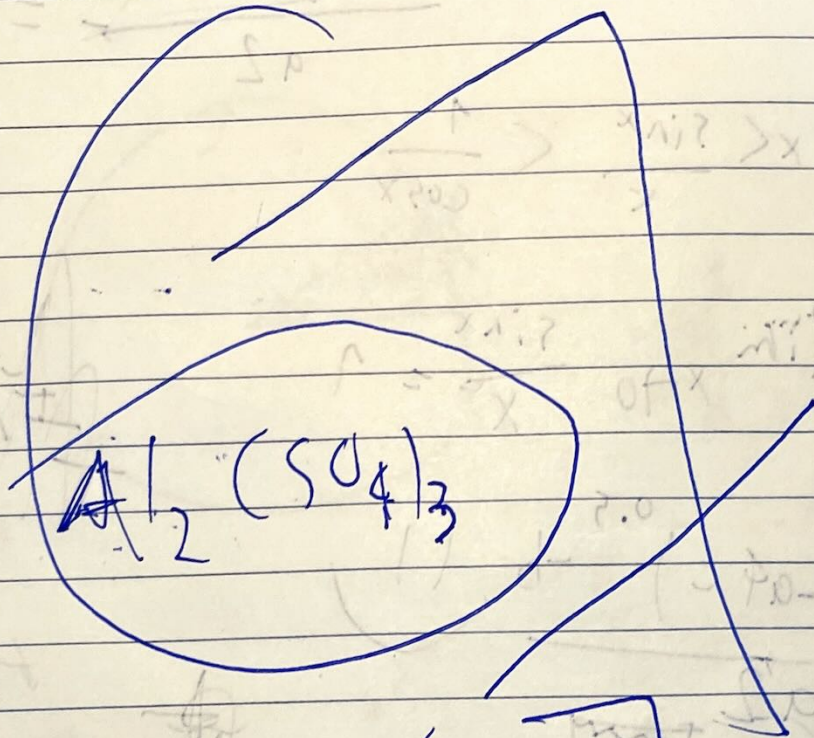
H₂O



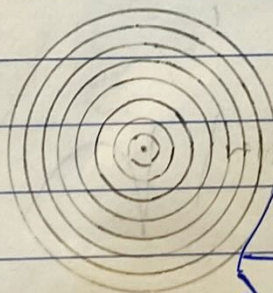
EXE



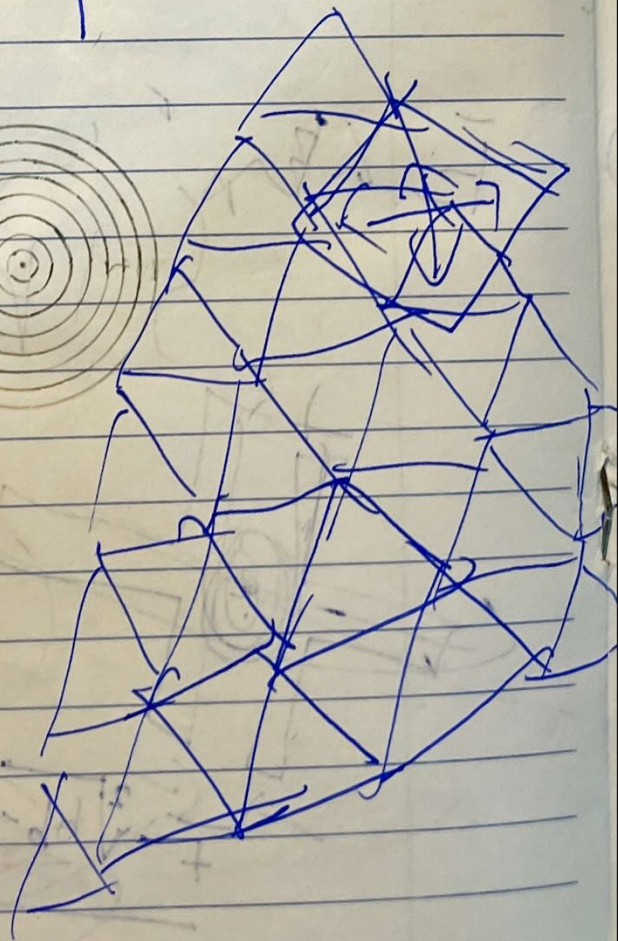
$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = x$$



$$\frac{- (b \pm \sqrt{b^2 - 4ac})^{0.5} - \frac{b}{2}}{a} = x$$



$$\frac{- (b \pm \sqrt{b^2 - 4ac})^{0.5} - b}{a^2} = x$$



ADD ADD

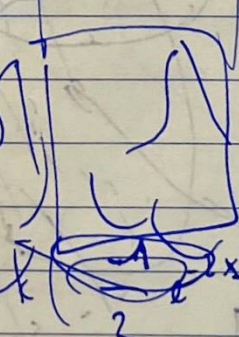
$$a \circ b = (b \circ a)$$

$$a \circ (b \circ c) = (a \circ b) \circ c$$

$$\frac{k}{2} \left(1 + \int_0^\infty e^{-2x} dx \right)$$

$$\frac{k}{2} \left(\frac{1}{2} + \frac{1}{2} \right) e^{-3x}$$

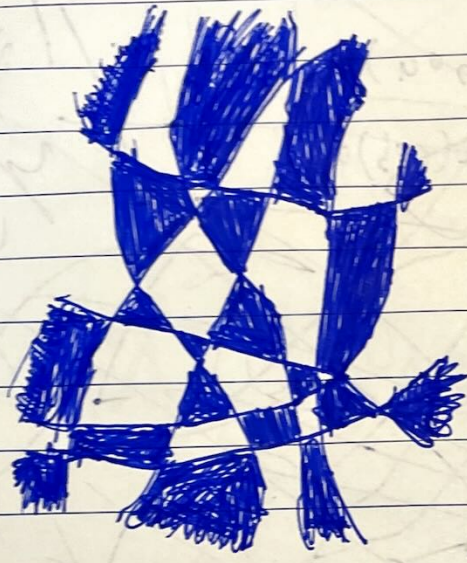
$$\frac{1}{2}$$



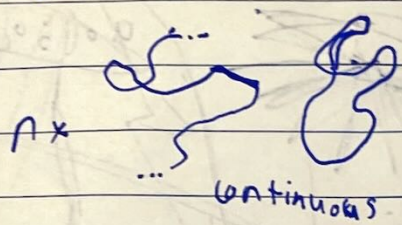
$$\int_0^\infty x e^{-2x} dx$$

$$x \left| \frac{-1}{2} e^{-2x} \right|$$

$$\frac{1}{2} k = 1$$

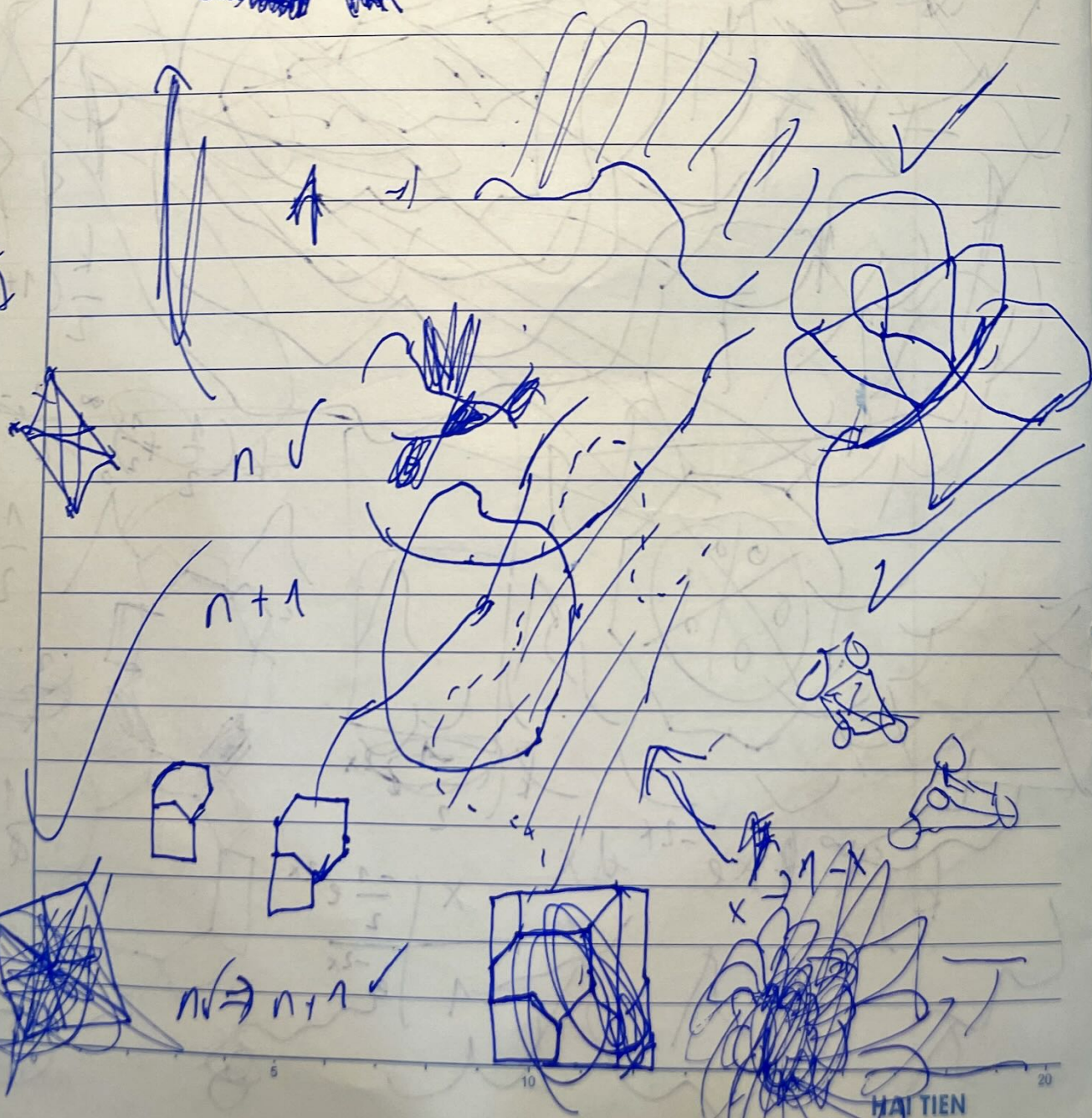


2. Theorem



0 → ✓

5



CONTINGENTS



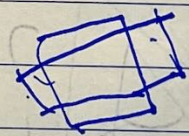
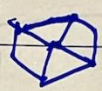
$$x^a \cdot \frac{x^{a+1}}{a+1}$$

$$x^a (\ln x)^n$$

$$\frac{1}{a+1}$$

$$\frac{x^{a+1}}{a+1}$$

$$\frac{(-1)^n n!}{(a+1)^{n+1}}$$



$$x^{a+1} \left(\frac{1}{a+1} \right)$$

$$x^{a+1} \left(\frac{\ln x}{a+1} \right)$$

$$\frac{1}{(a+1)^2}$$

$$x^{a+1} \left(\frac{(\ln x)^2}{a+1} \right)$$

$$\frac{2 \ln x}{(a+1)^2} + \frac{2}{(a+1)^3}$$

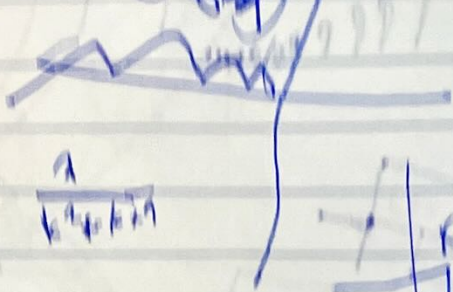
6

$$(ln x)^n$$

$$\frac{n(ln x)^{n-1}}{x}$$

6

$$p_n(x)$$

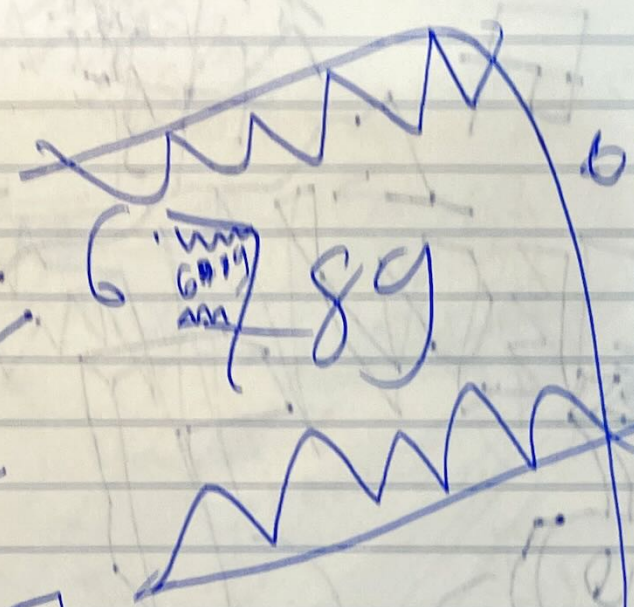


$$\frac{n}{k^2 + k + 1}$$

$$f(x) + \frac{20}{9n}$$

$$A_n = x(ln x)^n - n A_{n-1}$$

67



67

6

7

6

67

67

67

67

67

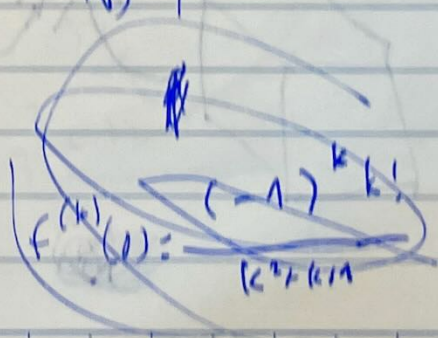
$$-1 \leq x \leq 1$$

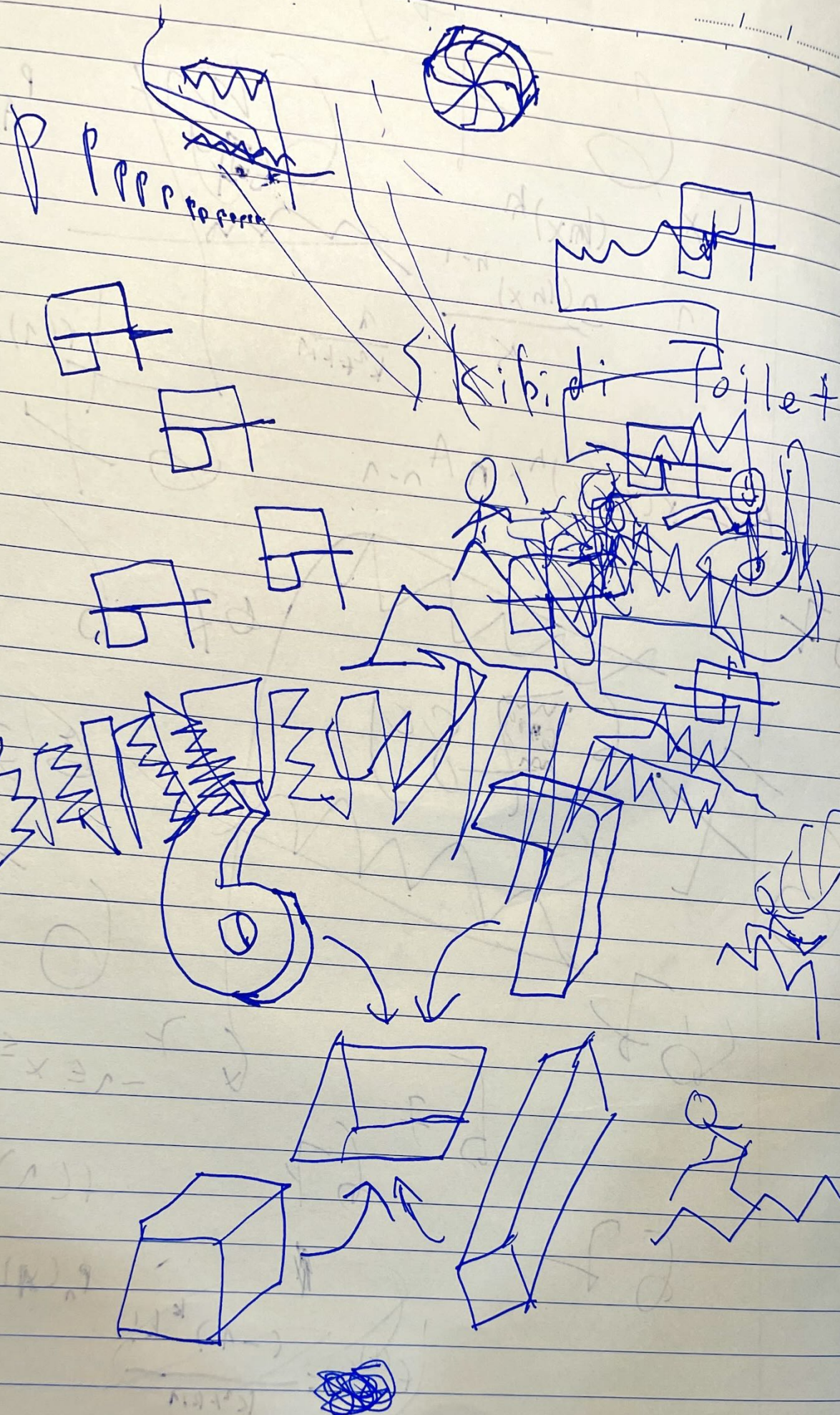
67

67

$$f(x)$$

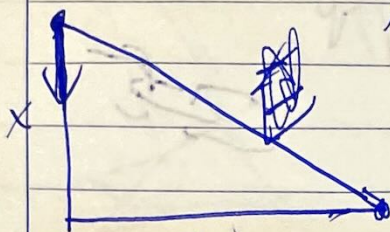
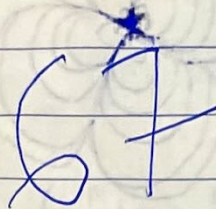
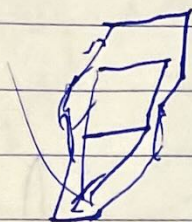
$$p_n(x)$$





MMMM

$$2 - \int_0^3 6 \cos(t + \sin t) dt$$



10

$$b = \sqrt{x^2 + 100}$$

$$\frac{dx}{dt} = 13$$

$$\frac{\ln n}{3^n n^2} x^n$$

$$\frac{\ln(n+1)}{3^{n+1} (n+1)^2} x^{n+1}$$

$$\frac{db}{dt} = \frac{db}{dx} \frac{dx}{dt}$$

$$3 \frac{\ln n (n+1)^3}{(n+1)^3}$$

$$\frac{1}{\sqrt{x^2 + 100}}$$

$$\frac{1}{1 + \sqrt{n}}$$

$$3 \cdot 2 \cdot 5 \cdot 2$$

$$\frac{dx}{dt} \frac{dx}{dt}$$

$$\frac{\ln n (n+1)^3}{\ln(n+1) n^3} \rightarrow 1$$

$$V = x^3 \sum \frac{1}{1+n}$$

$$\rightarrow \infty$$

$$\frac{2x^2}{2x+1}$$

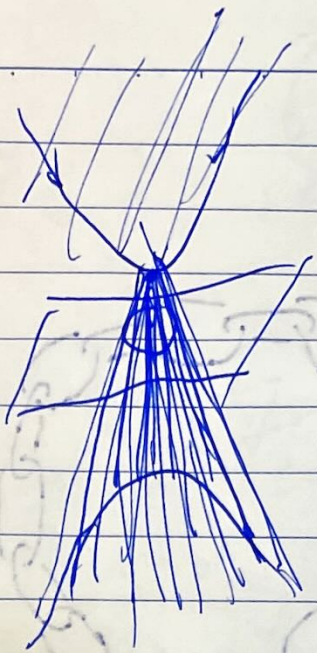
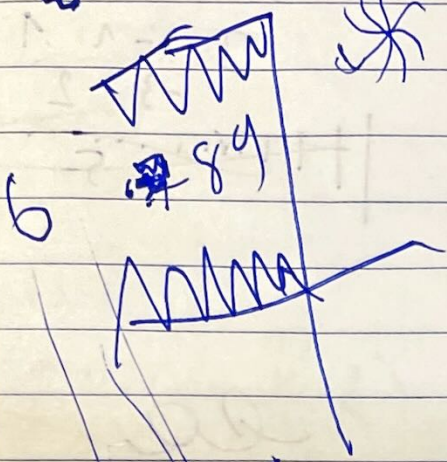
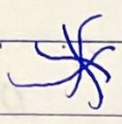
$$\int_0^1 \sqrt{(4e^{-t})^2 + (\sin(10\sqrt{t}))^2} dt$$

$$\frac{1}{2} + \lim_{x \rightarrow 0} \left(2x \sin \frac{1}{x} - \cos \frac{1}{x} \right)$$

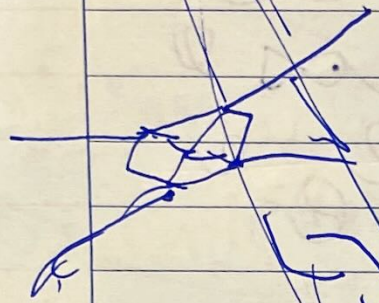
$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$\frac{x}{2} + x^2 \sin \frac{1}{x}$$

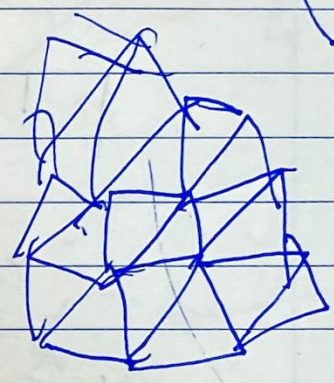
$$\lim_{x \rightarrow 0} \left(\frac{1}{2} + 2x \sin \frac{1}{x} - \cos \frac{1}{x} \right)$$



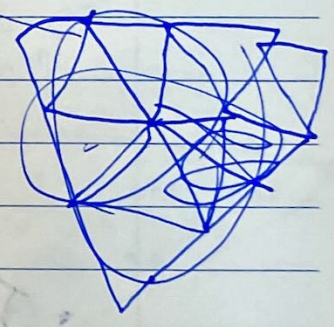
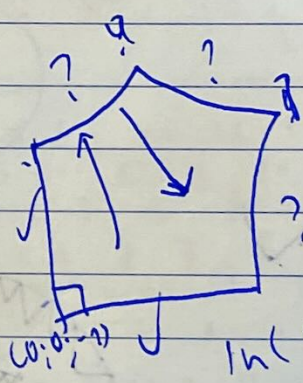
$$x^2 + y^2 - z^2 = 1$$



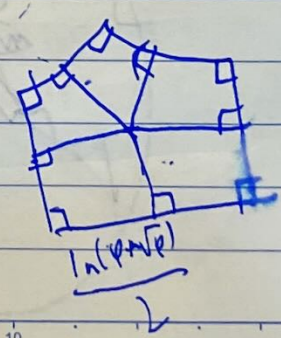
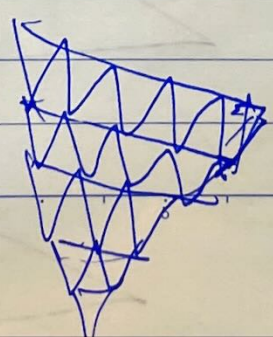
5, 4



C
H
A
I
N



$$\ln(\varphi + \sqrt{\varphi})$$



$$\ln(\varphi + \sqrt{\varphi})$$

1 1 2 3 5 8 13

0 1 1 2 3 5
1 0 1 2 3 5
-1 -1 0 1

2 a - 1 1 c - h c h c

-3 2

~~|||||~~

$ab + hca + c^2 \geq 0$

$(a+c)(b-c) \geq 0$

$(a+b)(a-b) \geq 0$

$(b+c)(a-c) \geq 0$

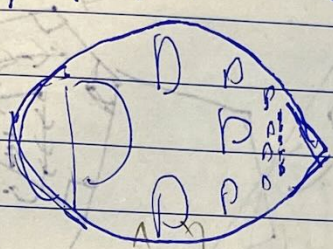
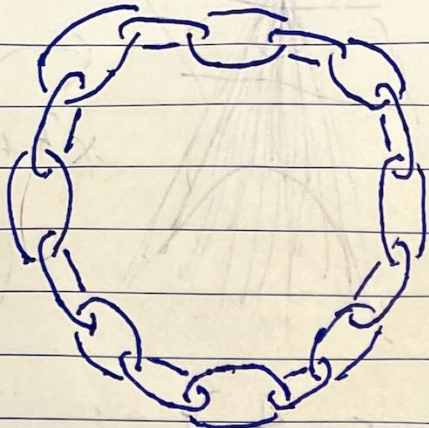
~~lo~~

$ab + hca - ca \geq c^2$

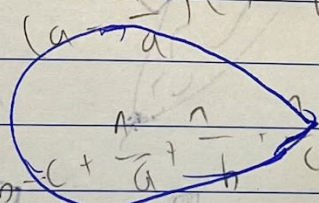
$ab + bc + ca \geq h^2$

$ab - bc + ca \geq c^2$

$2ab \geq b^2 + c^2$

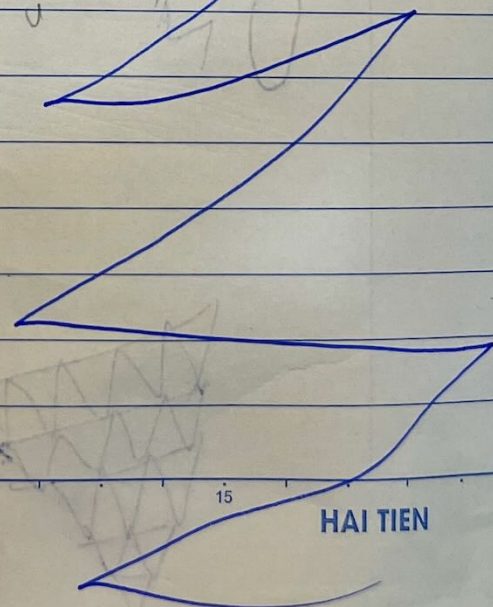
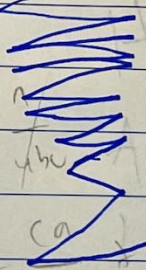


$(a - \frac{h}{a})(b - \frac{h}{b})(c - \frac{h}{c})$



$abc - a - b - c$

$ab - \frac{bc}{a}$



38

6/13

$f f f f$

abc

$$2c^2 + 2ca$$

$$a+b+c+a^2+b^2+c^2$$

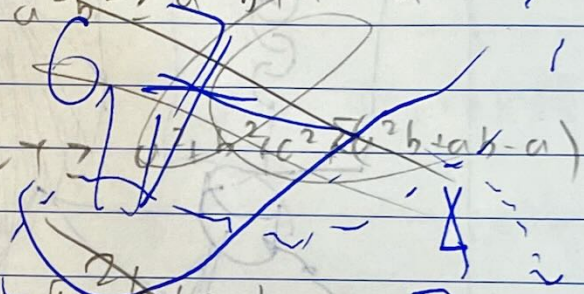
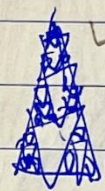
$$b(a-1)(ab-1) \geq 0$$

$f(x)$

$$a^2b^2 + a^2b + a^2 - a$$

$$(ab-1)(bc-1)(ca-1)$$

$$(a^2-1)(b^2(x)=f(2x))$$



$$-a^2bc - b^2ca - c^2ab$$

$$c^2b^2 + c^2a^2 + a^2b^2$$

$$a^2b^2 + b^2c^2 + c^2a^2 + ab^2bc + ca^2$$

$$a^2bc + b^2ca + c^2ab + a^2b^2 + b^2c^2 + c^2a^2$$

1 1 2 3 5 8 13 21 34 55 89 144 233

0 1 1 2 3 5 8 13 21 34 55 89

1 0 1 1 2 3 5 8 13 21 34

-1 4 0 1 4 2 3 5 8

(2, 5, 1, 1, 0, 1, 1, 2, 3)

-3 2 -1 1 0 1 1

5 -3 2 -1 1 0

(9-1) (-8, 5, 13, 2, -1)

13 -8 5 -3

$a^2b^2 + a^2b + a^2 - a$

-21 13 -8 1

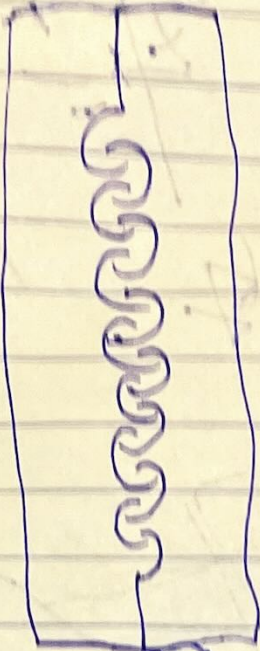
34 21

-99

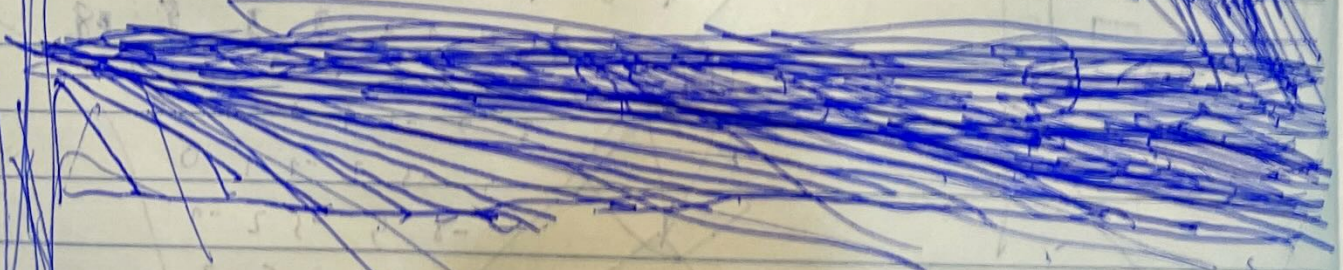
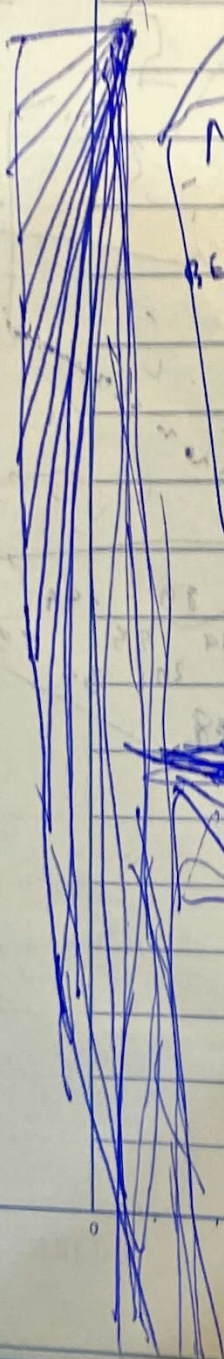
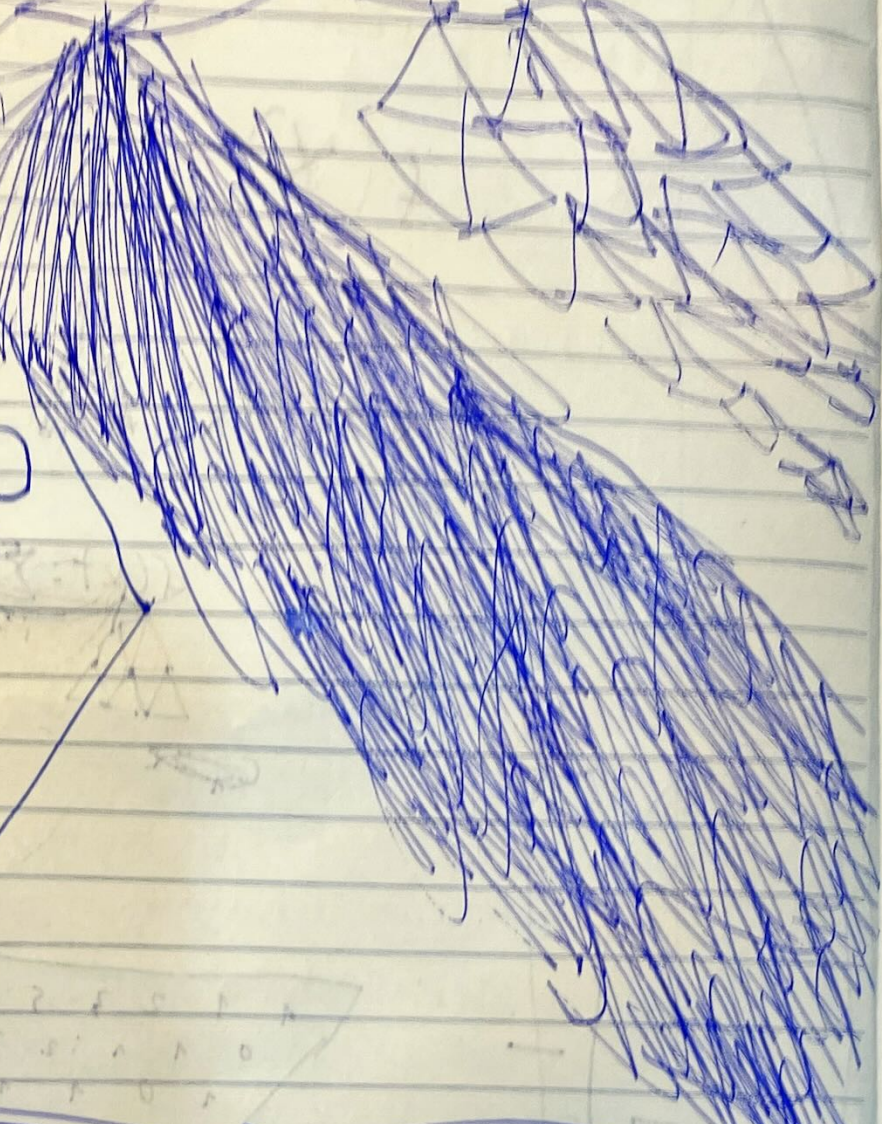
87

7

34



MILK
RES BY MARNO



MARNO

$$\frac{d}{dx} x f(x) = x \frac{d}{dx} f(x) + f(x) \quad X: f(x) \rightarrow x f(x)$$

$$\frac{d}{dx} x f(x)$$

$$f(g(x))$$

$$f \circ g$$

$$\frac{d}{dx} f(x)$$

$$f(x) + x f'(x)$$

$$f(x) + x \frac{d}{dx} f(x)$$

$$(f \circ g)(x) = f(g(x))$$

$$f \rightarrow f \circ \frac{d}{dx} \quad 67$$

$$f \circ g(x) \rightarrow f(g(x))$$

$$\frac{d}{dx} : g(x) \rightarrow g'(x)$$

$$f(g(x)) = g'(x)$$

$$\frac{d}{dx} f(x)$$

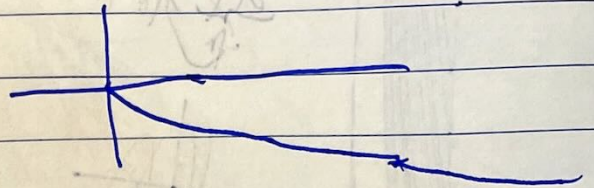
$$f: g(x) \rightarrow f(g(x))$$

$$f(g(x))$$

fo

$$f: g(x) \rightarrow f(x)g(x)$$

$$\frac{d}{dx} f(x)$$



$$\frac{d}{dx} f(g(x))$$

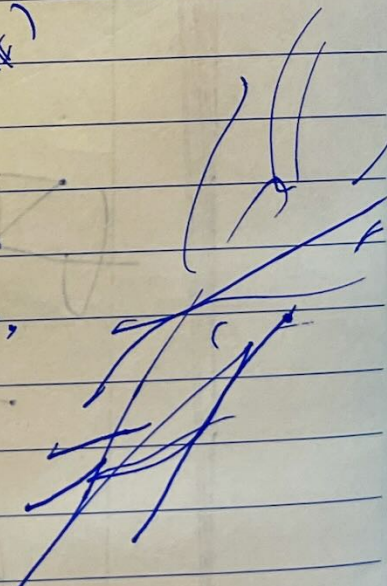
$$\frac{d}{dx} f(g(x))$$

$$\frac{d}{dx} f(x)g(x)$$

$$\frac{d}{dx} f(g(x))$$

$$f'(x)g(x) + f \frac{d}{dx} g(x)$$

$$\frac{d}{dx} f - f \frac{d}{dx} = f'$$



$$A' - \bar{A} = 0$$

$$A' = \bar{A}$$

$$A = \bar{A}$$

2025

20267

67

67

$$A = Ce^{x^2}$$

$$\frac{d}{dx} A = \bar{x} A$$

$$A' - \bar{x} A = 0$$

$$A: f(x) \rightarrow f(2x)$$

$$\frac{d}{dx} \frac{d}{dx} A = \bar{x} A$$

$$p_1 \rightarrow 10 \rightarrow x$$

$$A \frac{d}{dx} = \frac{d}{dx} A$$

$$A \frac{d}{dx} f(x)$$

$$A f(x)$$

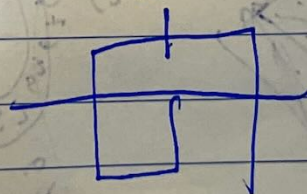
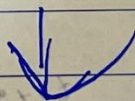
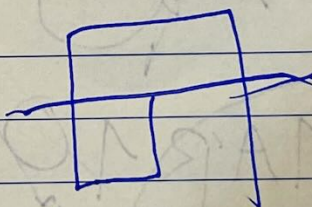
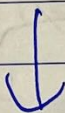
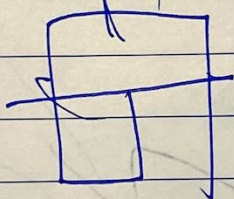
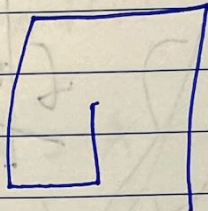
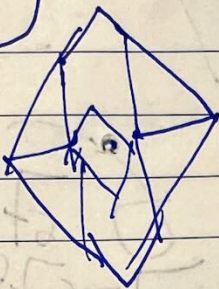
$$= A(x f(x)) - x A f(x)$$

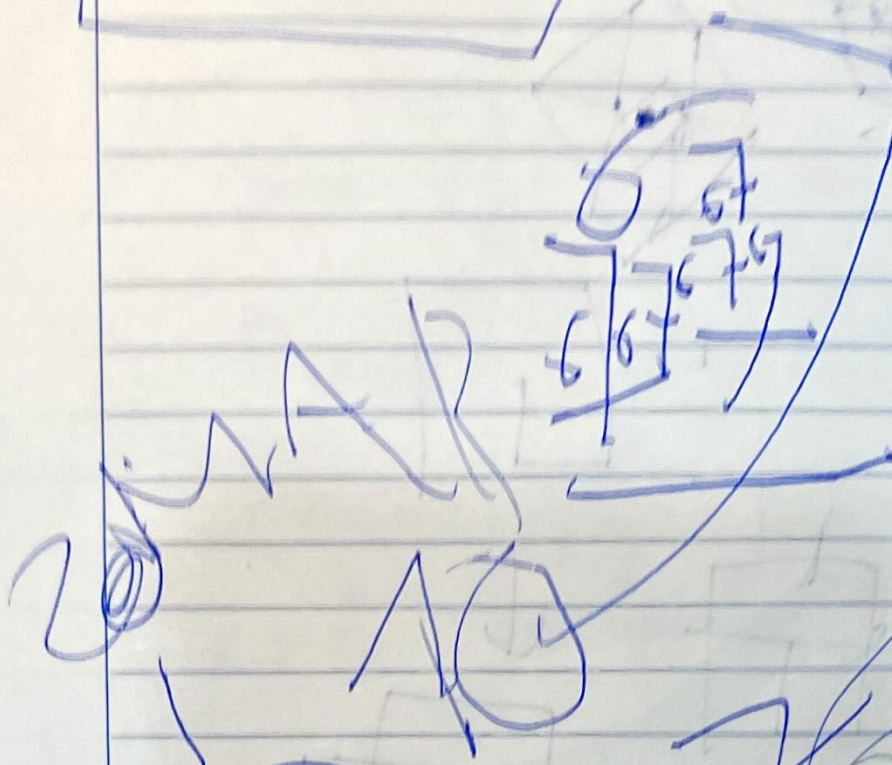
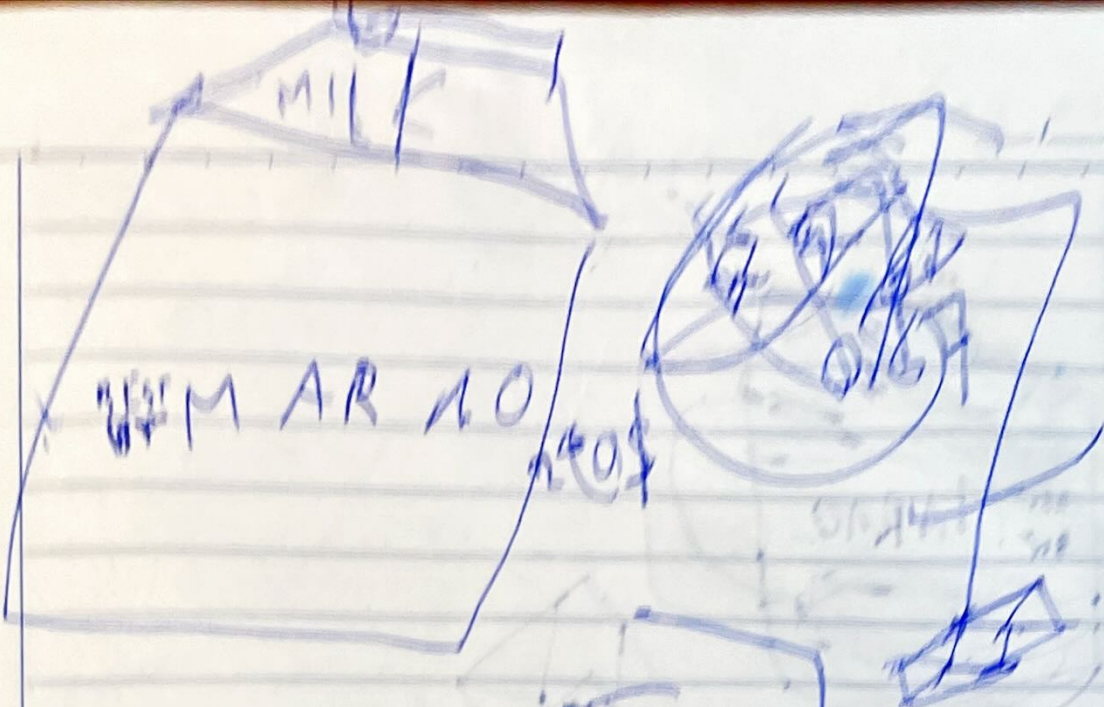
$$A(x f(x)) - x A f(x)$$

$$2x f(2x) - x f(2x)$$

$$A \frac{d}{dx} = x A$$

3





$$\prod \left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{5^2}\right) \left(1 - \frac{1}{7^2}\right) \dots$$

$$\frac{2}{1 \cdot 3} \cdot \frac{2 \cdot 4}{3^2} \cdot \frac{4 \cdot 6}{5^2} \dots$$

~~$$\sum_{n=1}^{\infty} \ln \left(1 - \frac{1}{n^2}\right)$$~~

$$\frac{6}{\pi^2}$$

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$$

~~$$\sin x = x \left(\frac{x^2}{\pi^2} - 1 \right) \left(\frac{x^2}{\pi^2} - 2^2 \right) \left(\frac{x^2}{\pi^2} - 3^2 \right) \dots$$~~

~~$$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$~~

~~x^4~~

~~$$\sin x = x \left(\frac{x^2}{\pi^2} - 1 \right) \left(\frac{x^2}{(2\pi)^2} - 1 \right) \left(\frac{x^2}{(3\pi)^2} - 1 \right) \dots$$~~

~~$$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$~~

~~x^4~~

$$\sin x = x \left(1 - \frac{x^2}{\pi^2} \right) \left(1 - \frac{x^2}{(2\pi)^2} \right) \left(1 - \frac{x^2}{(3\pi)^2} \right) \dots$$

$$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$x^1: 1 = 1 \quad \checkmark$$

$$x^3: \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots = \frac{\pi^2}{6} \quad \checkmark = A_1$$

~~x^5~~

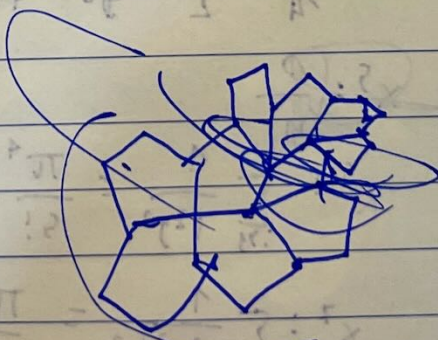
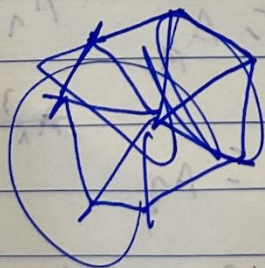
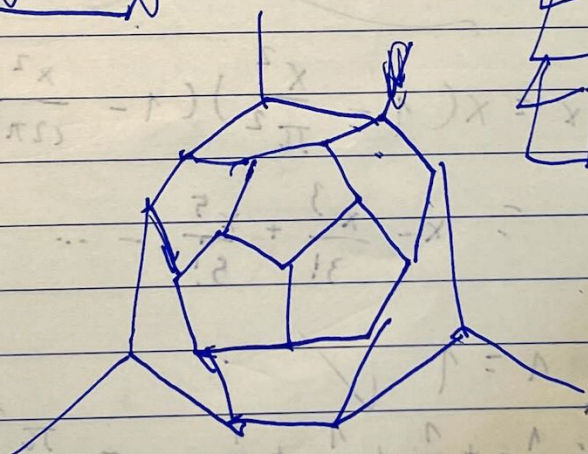
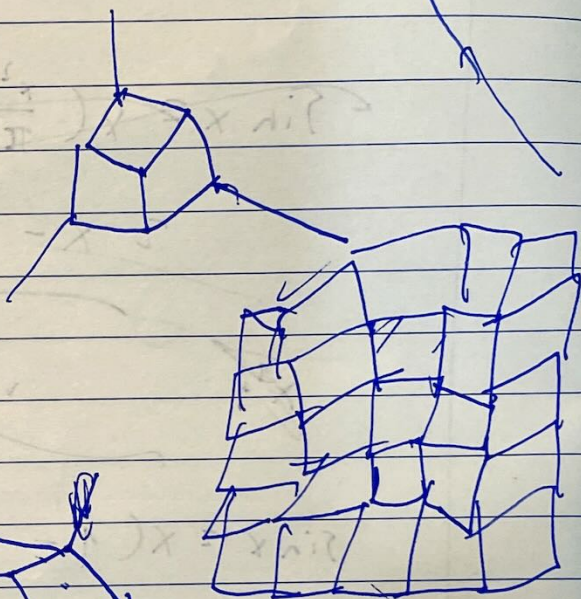
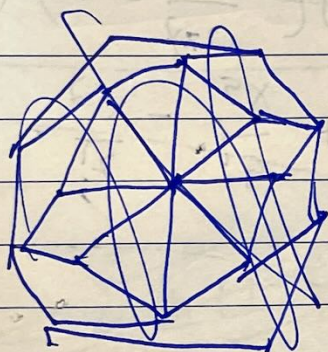
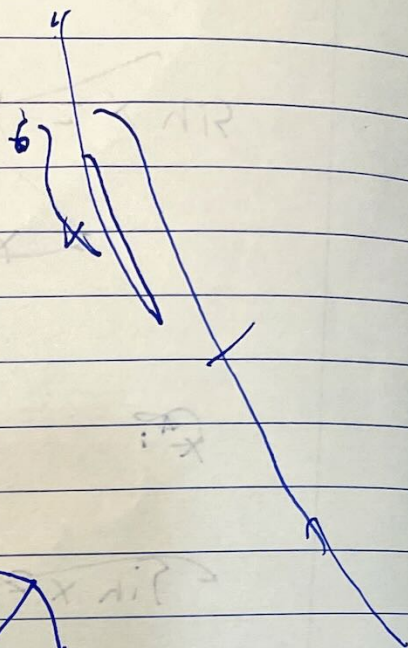
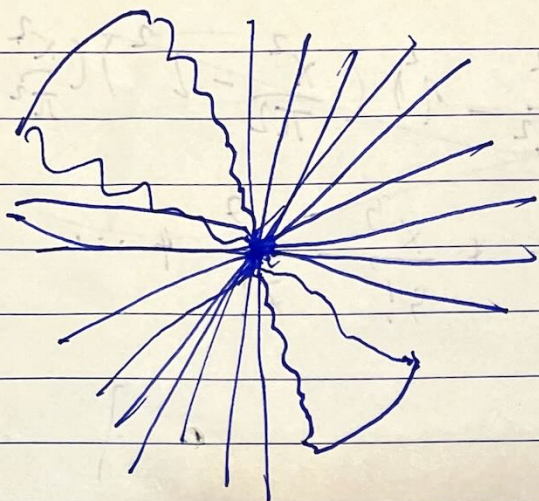
$$x^5: \sum_{j=1}^{\infty} \frac{1}{12j^2} = \frac{\pi^4}{5!} = A_1 A_2$$

$$A_1^2 - 2A_2 = \sum \frac{1}{n^4} = \frac{\pi^4}{90}$$

~~$$A_1^3 - 6A_3$$~~

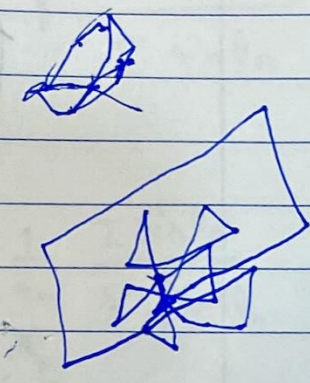
$$x^7: \sum_{j=1}^{\infty} \frac{1}{2 \cdot j^2 k^2} = \frac{\pi^6}{7!} = A_3$$

$$x^{2n+1}: \sum_{j=1}^{\infty} \frac{1}{i_1^2 \cdot i_2^2 \dots i_n^2} = \frac{\pi^{2n}}{(2n+1)!} = A_n$$

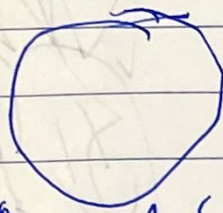
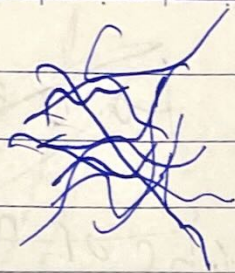




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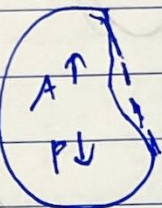
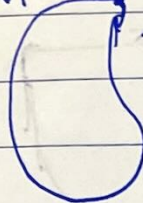


Max $\frac{A}{p^2}$?

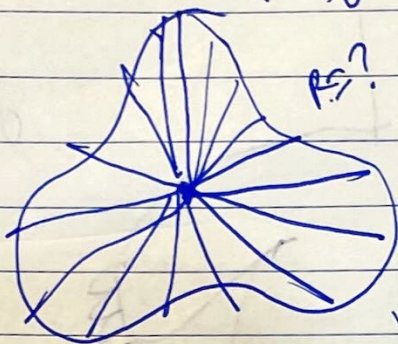
$$A = \int_0^{2\pi} \pi r^2 \frac{d\theta}{2\pi} = \frac{1}{2} \int_0^{2\pi} r^2 d\theta$$

$r(\theta)$

p ?



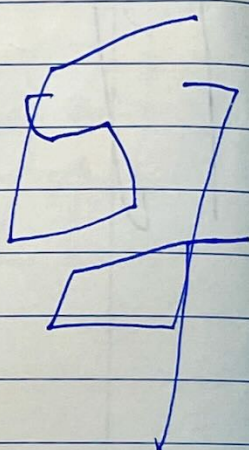
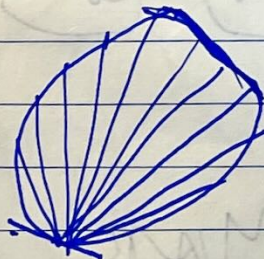
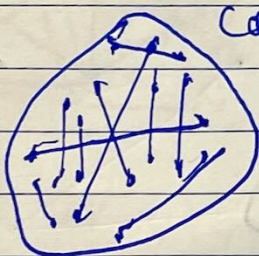
pppppppp
p
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Max $\frac{A}{p^2}$

p ?

convex



$$\sim 3x^2 + 2x - 1$$

$$\sim 2x^2 + 2 \quad \sim x^2 + 2x - 1$$

$$x^2 - 2x + 1$$

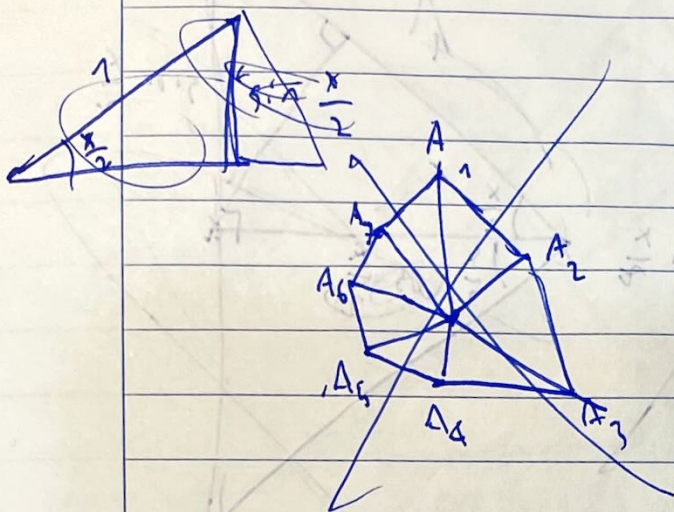
$$\sim 2(x+1)(x-1) - (x-1)^2$$

$$\text{Max } n \quad \frac{A}{p^2} \leq \frac{1}{8n} \frac{\sum \sin \theta_n}{\sum (\sin \frac{\theta_n}{2})^2}$$

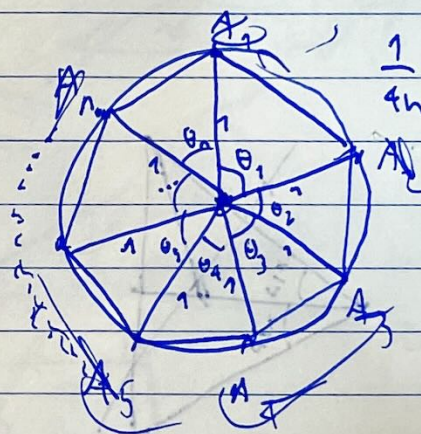
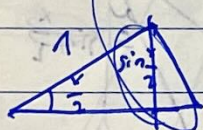
$$\frac{1}{6n} \frac{\sum \sin \theta_n}{\frac{n-1}{2} \sum \cos \theta_n}$$

$$\frac{A}{p^2} = \frac{1}{8} \frac{\sum \sin \theta_n}{(\sum \sin \frac{\theta_n}{2})^2}$$

$$\frac{p^2}{A} = 8 \frac{(\sum \sin \frac{\theta_n}{2})^2}{\sum \sin \theta_n}$$



$$\left(\sin \frac{\theta_n}{2} \right)^2 = ?$$



$$\frac{1}{4n} \frac{\sum \sin \theta_n}{1 - \sum \cos \theta_n}$$

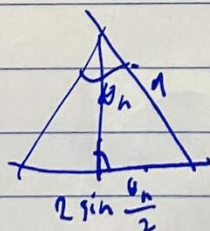
$$\frac{\sin \frac{2\pi}{n}}{1 - \cos \theta_n}$$

$$\frac{\sin \theta_n}{2}$$



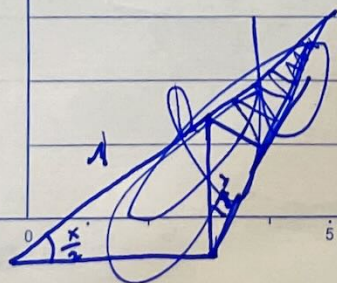
$$\text{Max } \frac{A}{p^2} \Rightarrow A_1 A_2 A_3 \dots A_n A_n$$

$$\text{max } \frac{A}{p^2} \Leftrightarrow \theta_1 = \theta_2 = \dots = \theta_n$$



$$p = \sum 2 \sin \frac{\theta_n}{2} = 2 \sum \sin \frac{\theta_n}{2} \quad \frac{1}{4n} \frac{\sin \frac{2\pi}{n}}{1 - \cos \frac{2\pi}{n}}$$

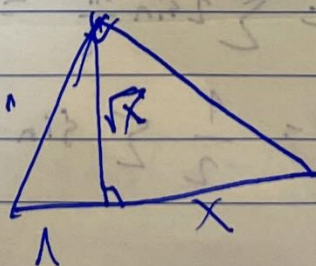
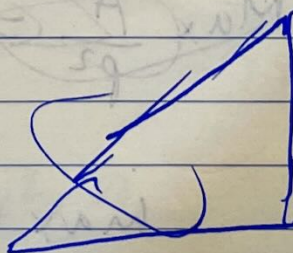
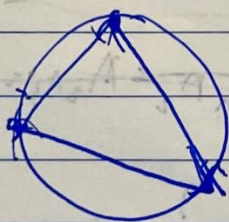
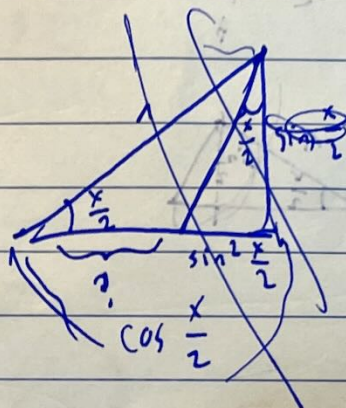
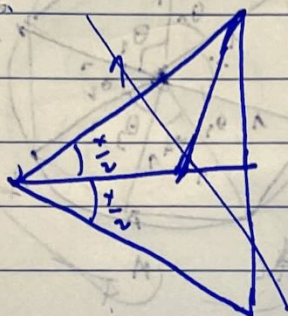
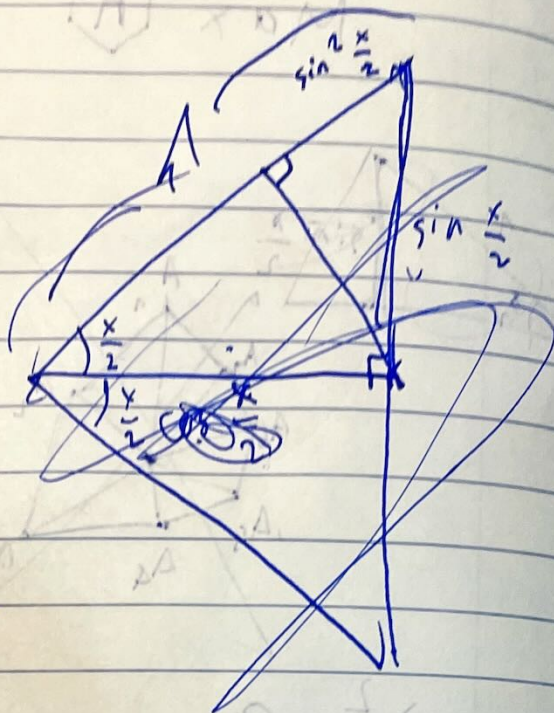
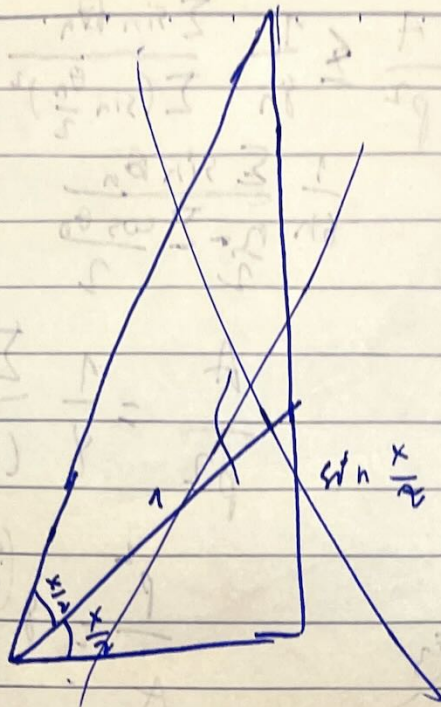
$$A = \frac{1}{2} \sum \sin \theta_n$$



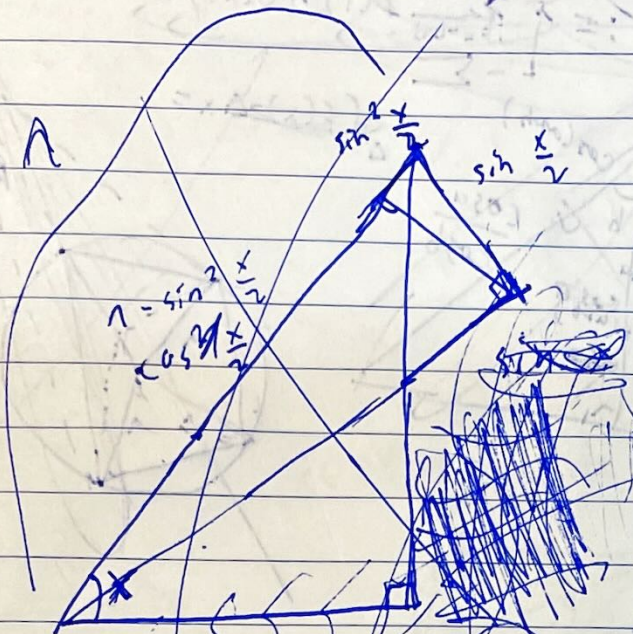
$$2 \pi \sin \frac{\pi}{n}$$

$$\frac{\frac{n}{2} \sin \frac{2\pi}{n}}{(2 \sin \frac{\pi}{n})^2}$$

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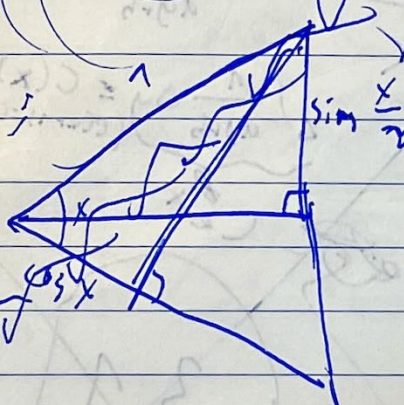
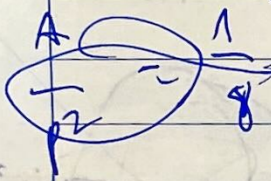


$$\tan^2 \frac{x}{2} = \frac{1 - \cos x}{1 + \cos x}$$



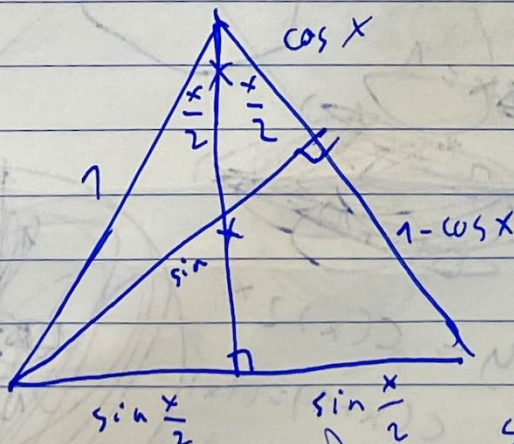
$$\cos^2 \frac{x}{2} = 1 - \frac{1 - \cos x}{2}$$

$$\sin^2 \frac{x}{2} = \frac{1 - \cos x}{2}$$



$$2 - 2\cos x = 4 \sin^2 \frac{x}{2}$$

$$\sin^2 x + (1 - \cos x)^2 = 4 \sin^2 \frac{x}{2}$$

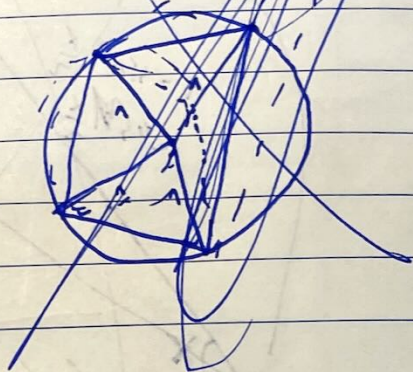
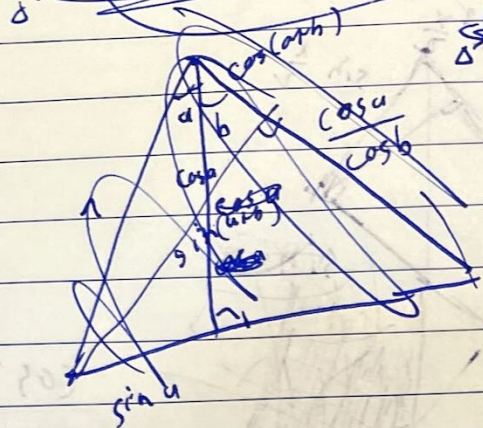


$$\sin^2 x + \cos^2 x + 1 - 2\cos x = 4 \sin^2 \frac{x}{2}$$

$$\frac{\Delta f(x)}{\Delta x} := f(x+1) - f(x)$$

$$\Rightarrow \int_a^b \frac{\Delta f(x)}{\Delta x} \Delta x = f(b) - f(a) \quad (b-a \in \mathbb{Z})$$

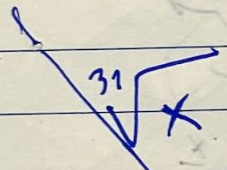
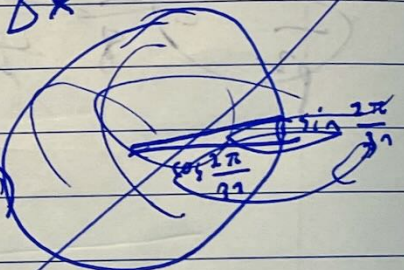
$$\int_a^b f(x) \Delta x := \sum_{i=a}^{b-1} f(i) \Delta x$$



$$\frac{\Delta y}{\Delta x} = ay + b \quad x, y \in \mathbb{N}$$

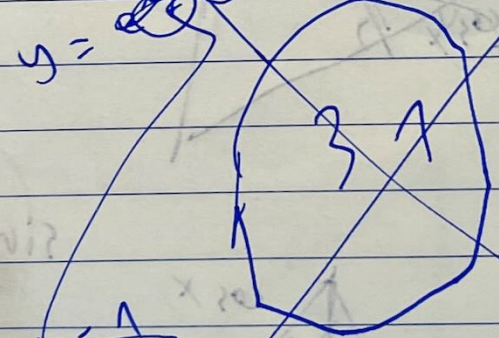
$$\frac{1}{ay+b} \Delta y = \Delta x$$

$$\Delta \int \frac{1}{ay+b} \Delta y = C(x) \quad (C(x+1) = C(x))$$

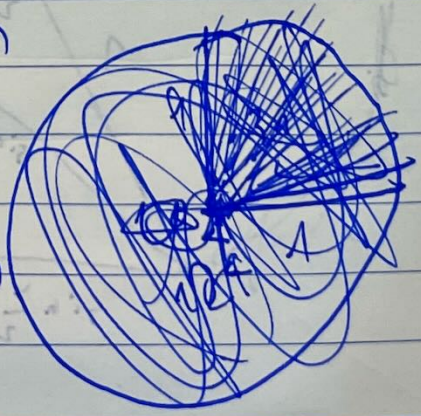


$$\frac{dy}{dx} = y$$

$$\Delta y = y$$



$$y = C(x)2^x \quad C(x+1) = C(x)$$



$$(a+b)^2 = \cancel{a^2}$$

$$a^2 \leq x \leq a \max$$

$$-2ab + b^2 = x - a^2$$

$$b(2a + b)$$



~~$$\int_{\Delta} f(x) dx = \sum_{i=1}^n$$~~

$$x \in \mathbb{N} \Rightarrow \int_0^x f(x) dx = \sum_{i=0}^x f(i) + C(x)$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

A cycle

$$\int_{\Delta} f(x) \Delta x := \sum_{i=0}^{n-1} f(i \cdot \Delta x) \cdot \Delta x$$

$$\sum_{i=0}^n f(i) = - \sum_{i=1}^{n-1} f(-i)$$

$$n = 4$$

~~$$\int f(x) \Delta x$$~~

Handwritten notes on lined paper:

10

$\sqrt{10}$

100 101 10

100 100

10000

1001

111100

10101

101100

101100

10110000

10110000

101100

$$\sum_{i=1}^{-1} f(i) = ?$$

$$\sum_{i=1}^n f(i) = f(1)$$

$$\sum_{i=1}^n f(i) = f(n) + \sum_{i=1}^{n-1} f(i)$$

$$\sum_{i=0}^{-n} = \underbrace{\sum_{i=0}^{-1}}_{=0} f(i) - f(-1) - f(-2) - \dots - f(-n+1)$$

$$\sin \frac{2\pi}{17}$$

$$e^{\frac{2i\pi}{17}}$$

$$\cos \frac{2\pi}{17} + i \sin \frac{2\pi}{17}$$

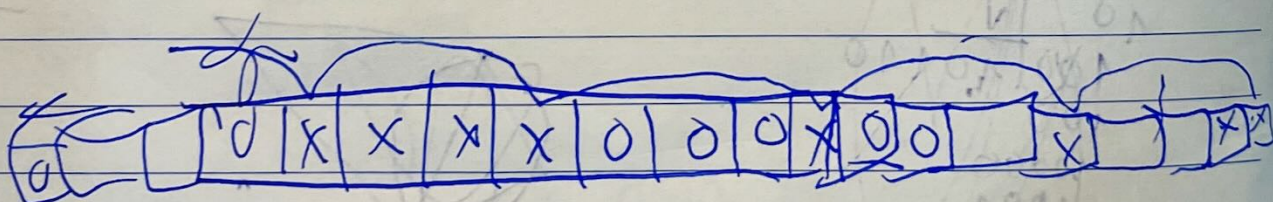
$$x^{17} = 1$$

$$2^{2^n} + 1$$

$$x^2 - 2x + 2$$

$$\sqrt{1 - \sin^2 \frac{2\pi}{17}}$$

$$2^{2^5} + 1 = 641 \cdot 6700417$$



17

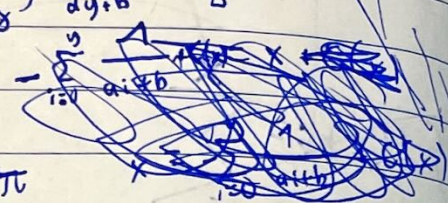
360

17

$$\frac{dy}{dx} + ay + b = 0$$

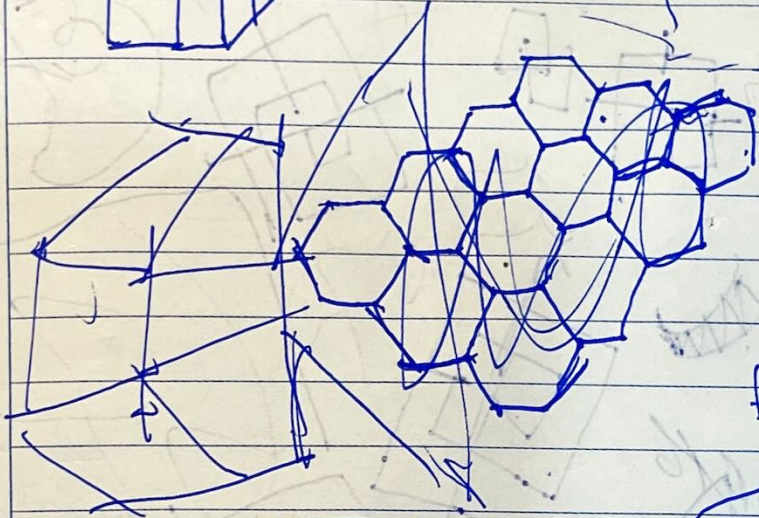
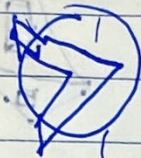
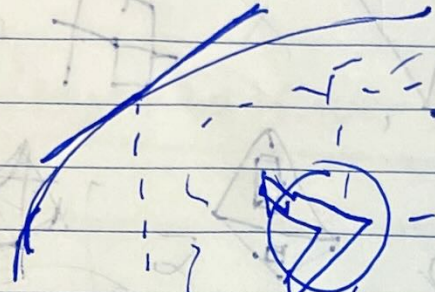
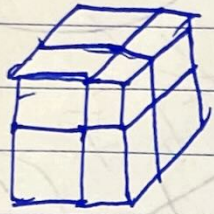
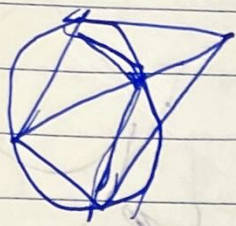
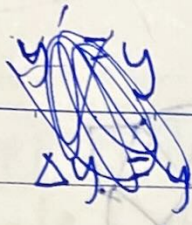
$$(ay + b) \Delta y =$$

$$\int \frac{-1}{ay+b} \Delta y \approx \int \frac{-1}{ay+b} \Delta x$$



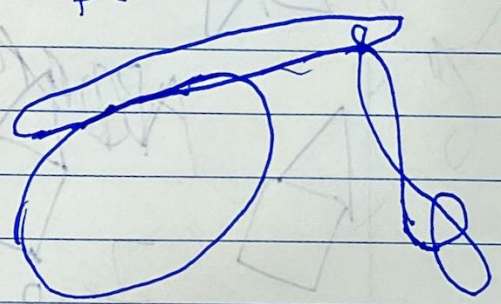
17

~~x, y \in \mathbb{N}~~



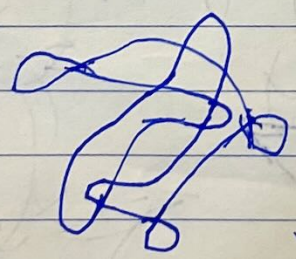
$$f(x) - f(a) \approx (x-a)f'(a)$$

$$\sqrt{x} - \sqrt{a} \approx \frac{x-a}{2\sqrt{a}}$$

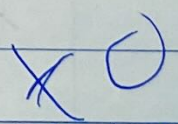
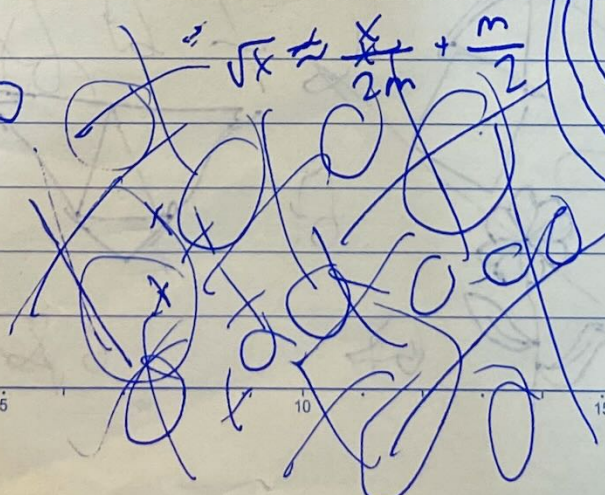


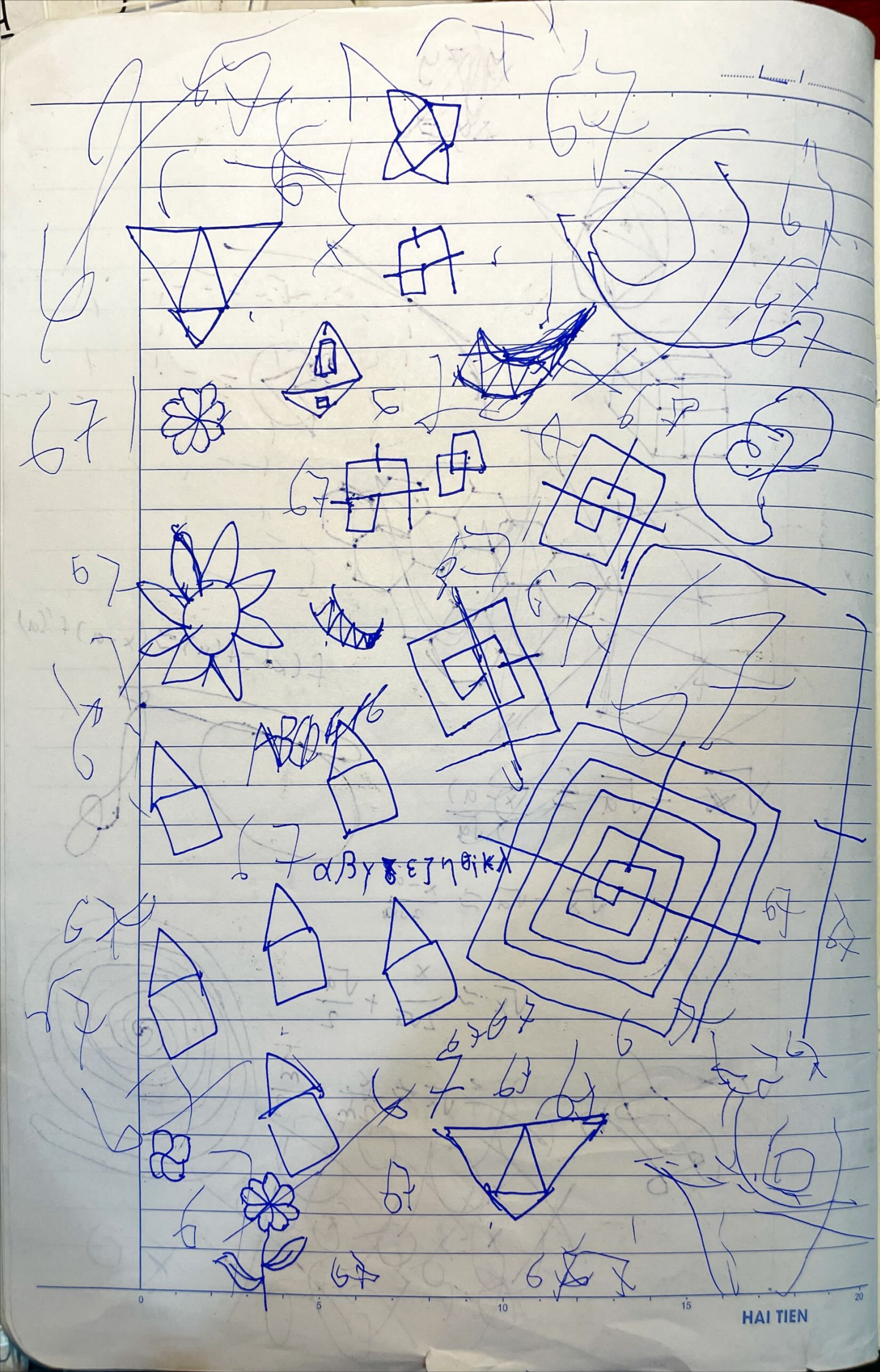
$$\sqrt{x} - \sqrt{a} \approx \frac{x-a}{2\sqrt{a}}$$

$$\sqrt{x} \approx \frac{x}{2\sqrt{a}} + \frac{\sqrt{a}}{2}$$



$$\sqrt{x} \approx \frac{x}{2\sqrt{a}} + \frac{\sqrt{a}}{2}$$





$$\text{Happiness} = f(t) \quad f(0) = 0$$

$$\text{People} = p(t) \quad \text{Sleep} = r(t)$$

$$\text{Alone activities} = q(t)$$

$$\frac{df}{dt} = \frac{(1 - p^2 + 3q)(100 - |f|)}{100} \quad \leftarrow rf$$

α/β γ δεζη θικλ